

DataNova

AI-Powered Business Intelligence & Predictive Analytics Platform

DataNova transforms raw business data into meaningful analytics, predictive models, and AI-powered business insights.

The platform combines **Data Engineering, Data Analytics, Machine Learning, and Generative AI** into a unified system. Users can upload business datasets, process them through automated ETL pipelines, store structured data in PostgreSQL, manage files through MinIO, visualize business performance, generate predictions, and interact with their data using an AI assistant.

Core Capabilities

-  Interactive Business Intelligence Dashboard
-  CSV and Excel Data Ingestion
-  Automated ETL & Data Quality Processing
-  PostgreSQL Data Management
-  MinIO Object Storage
-  Business Analytics & Visualization
-  AI-Powered Business Assistant
-  Sales & Demand Forecasting
-  Customer Segmentation & Churn Prediction
-  Anomaly Detection
-  Automated PDF & Excel Reports
-  Authentication & Role-Based Access
-  Docker-Based Development & Deployment

Technology Stack

Frontend: Next.js, React, TypeScript, Tailwind CSS

Backend: Python, FastAPI

Database: PostgreSQL

Datastore: MinIO

Data Processing: Pandas, NumPy

Machine Learning: Scikit-learn, XGBoost, Prophet

AI: Gemini API

Visualization: Recharts

DevOps: Docker, GitHub Actions

DataNova — Transforming Data into Intelligent Decisions.